DUPIRE'S LOCAL VOLATILITY MODEL

A local volatility model provides a deterministic extension of the classical Black–Scholes model.

4.1 Asset Dynamics

Under the risk neutral measure, the asset price is assumed to follow a diffusion process with volatility specified as a deterministic function of both time and current value of the underlying asset (*i.e.,* follows a GBM with local volatility):

$$\frac{dS}{S} = rdt + \sigma_l dX. \tag{14}$$

4.2 Derivation of Local Volatility from Dupire's Equation

Dupire, in 1994, derived an explicit equation for the local variance in terms of European call option prices. If q is the dividend yield, and σ_l is the local volatility. Here is Dupire's equation in most general form:

$$\frac{\partial C}{\partial T} = \frac{1}{2}\sigma_l^2 E^2 \frac{\partial^2 C}{\partial E^2} + (r-q)\left(C - E\frac{\partial C}{\partial E}\right) - rC. \tag{15}$$

The derivation of Dupire's equation can be completed via the Fokker-Planck equation [Risken 1996]. To solve for the local variance in (15), we simply rearrange to obtain:

$$\sigma_l^2(E,T) = \frac{\frac{\partial C}{\partial T} + (r-q)E\frac{\partial C}{\partial E} + qC}{\frac{1}{2}E^2\frac{\partial^2 C}{\partial E^2}}. \tag{16}$$

Note that rC is not represented in (16) since the equation is derived under the assumption that the call price is a forward price. Once you consider and write the option as a forward price, you remove the discount factor (the term which represents the call option's value back to present). The Dupire equation ensures that the local volatility surface is the unique deterministic function consistent with observed European option prices, provided sufficient smoothness and no-arbitrage conditions hold.

After finding the implied volatility from the Black-Scholes model, we can find the call price using the implied volatility. We can compute Θ and the partial derivatives of C with respect to E, the strike price. Using this, we can plug these values into Dupire's local volatility equation:

$$\sigma_l = \sqrt{\frac{\Theta + (r-q)\,E\frac{\partial C}{\partial E} + qC}{\frac{1}{2}E^2\frac{\partial^2 C}{\partial E^2}}}. \tag{17}$$

4.3 Plotting the Local Volatility Surface

We can find the local volatility values for all strikes and maturities for European call options.

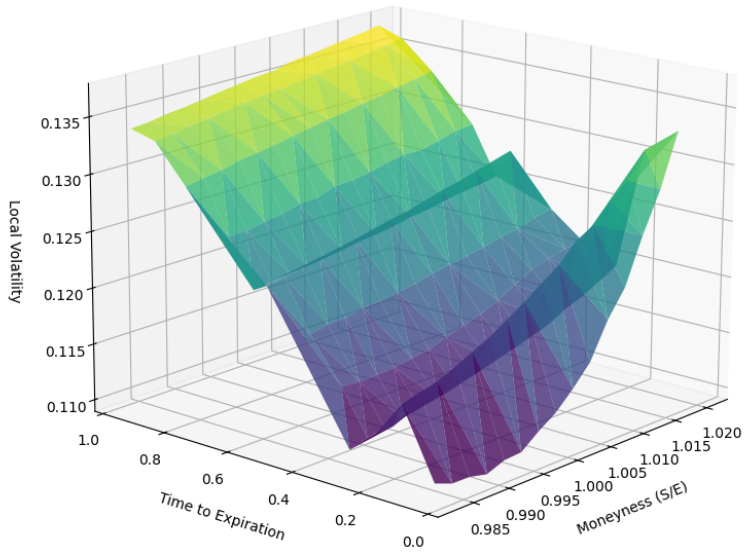


Figure 1: Local Volatility Surface.

Figure (1) displays the Dupire output as a result of running the Python code for the dataset.

4.4 Mathematical Properties of the Dupire model

The Dupire local volatility model has many mathematical features: once calibrated, volatility becomes a deterministic function of price and time, making the model complete for European options and allowing pricing to be reduced to a one dimensional PDE while achieving an exact fit to the implied volatility surface. However, its deterministic nature prevents it from capturing key market behaviours such as volatility clustering (the observation that large movements in an underlying's price tend to be followed by further large movements in price, and small price movements tend to be followed by small movements) and variance risk premia (which represents the difference between the market's implied and realised volatility). Hedging strategies derived from local volatility may therefore misrepresent exposure to future volatility shocks, particularly for path-dependent (contracts whose payoff depends not just on the final price of the underlying asset at maturity, but on the entire path the asset takes over time) or exotic (beyond vanilla) derivatives. Computing the local volatilty requires differentiating call prices with respect to strike and maturity, which can introduce numerical instability when market data is sparse or noisy. Degeneracy can also occur at extreme strikes or short maturities, which raises concerns about stability and arbitrage consistency.